

CIS 371 Web Application Programming

HTTP



Lecturer: **Dr. Yong Zhuang**

Practice: Event Handling

Counting Click

Answer



Given the HTML and JavaScript code in the above link, implement functionality for the `<h1 id="h1counter">` element so that when it is clicked, the `<p id="outputH1counter">` displays 'Clicked __ times'.

HTTP

- HyperText Transfer Protocol
- Invented by Tim Berners Lee @ CERN
- A protocol for delivering resources over the web
- TCP/IP connections, default (server) port 80
- HTTP client & HTTP server



Other network Transfer Protocols

- FTP: File Transfer Protocol
- FTPS: Secure FTP
- SMTP: Simple Message Transfer Protocol
- NTP: Network Time Protocol

Why learn the details of HTTP?



Why learn the details of HTTP?

HTTP requests from your program

- To understand how your program sends and receives HTTP requests.
- To debug and optimize HTTP requests from your program.
- To design better APIs and handle client–server communication effectively.

Web Client/Server Architecture

(3) Send “contents” (HTML + CSS + JS & **other data**)

(4) Present “contents”

(5) Run Code

(6) Handle user input

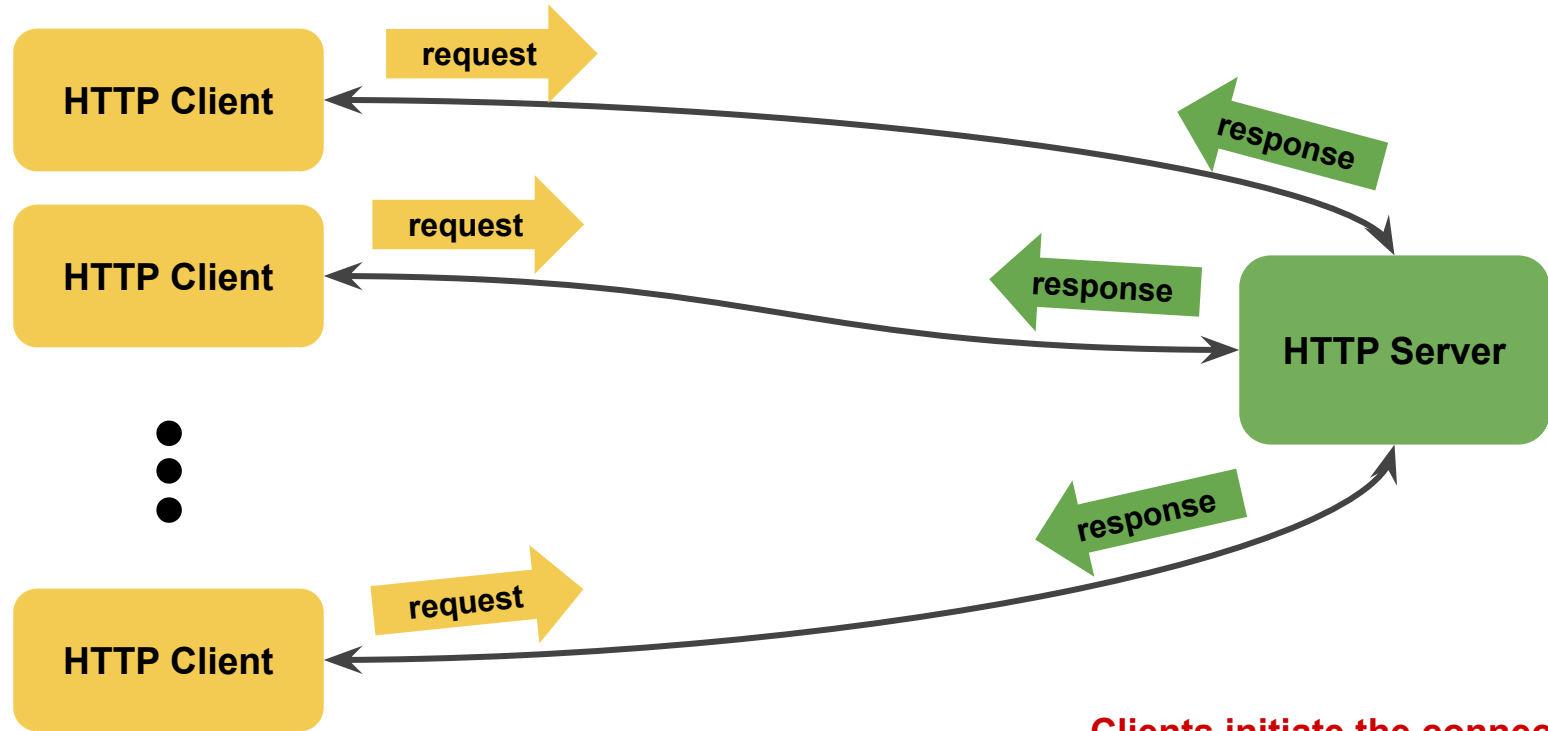


(2) Run Code



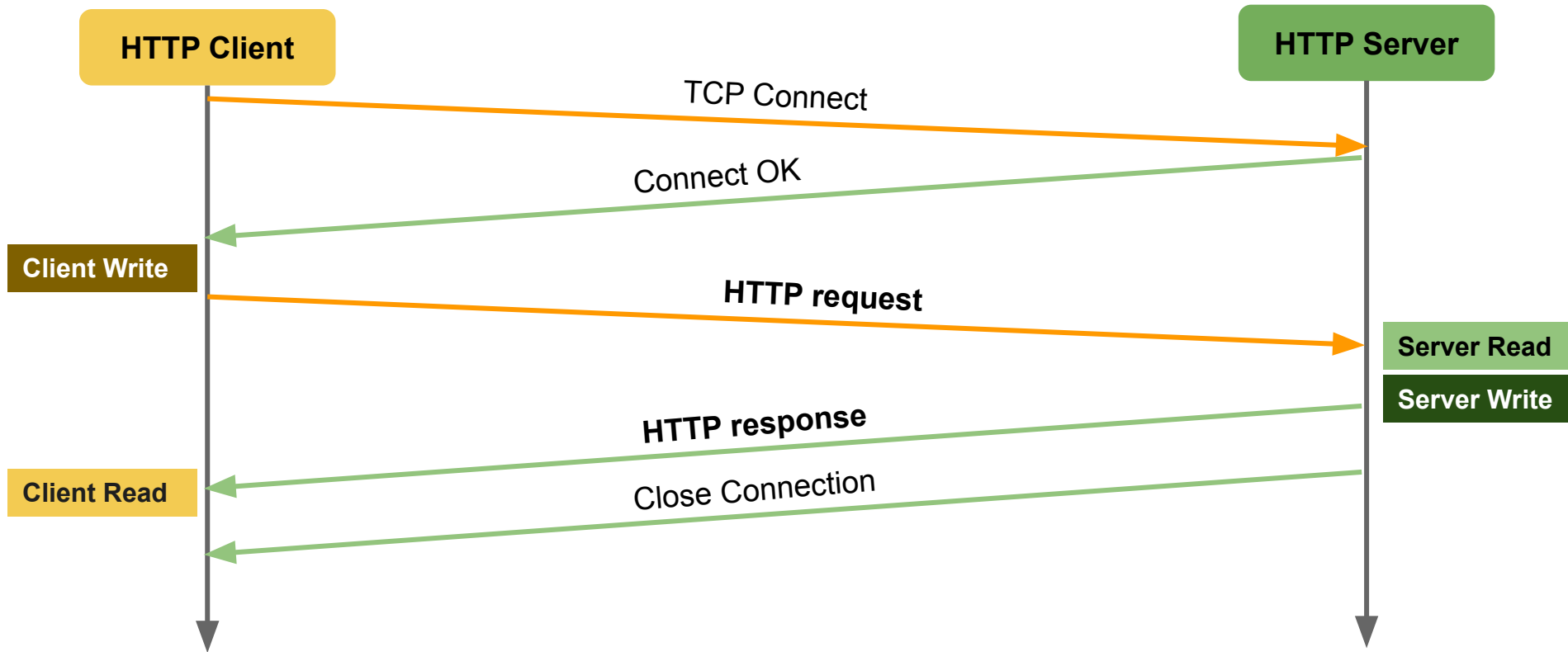
(1) Send “user input”

HTTP Communication Model



Clients initiate the connection!!!

Transaction Timeline (TCP Sockets)

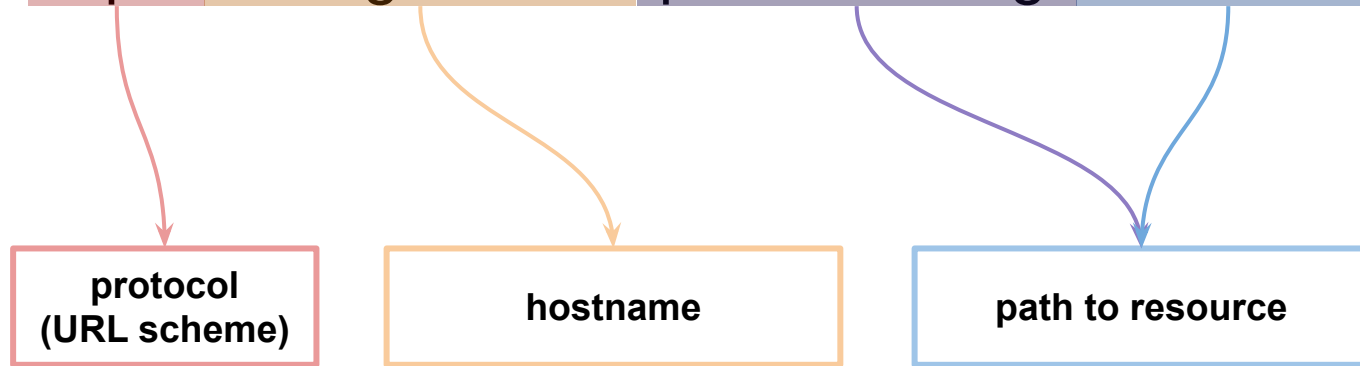


HTTP URL: Uniform Resource Locator

http:// www.gvsu.edu /files/registrar/622GX/7155/ admission.pdf

http:// www.gvsu.edu /files/img/article/frontpag/ 5123FG73A.jpg

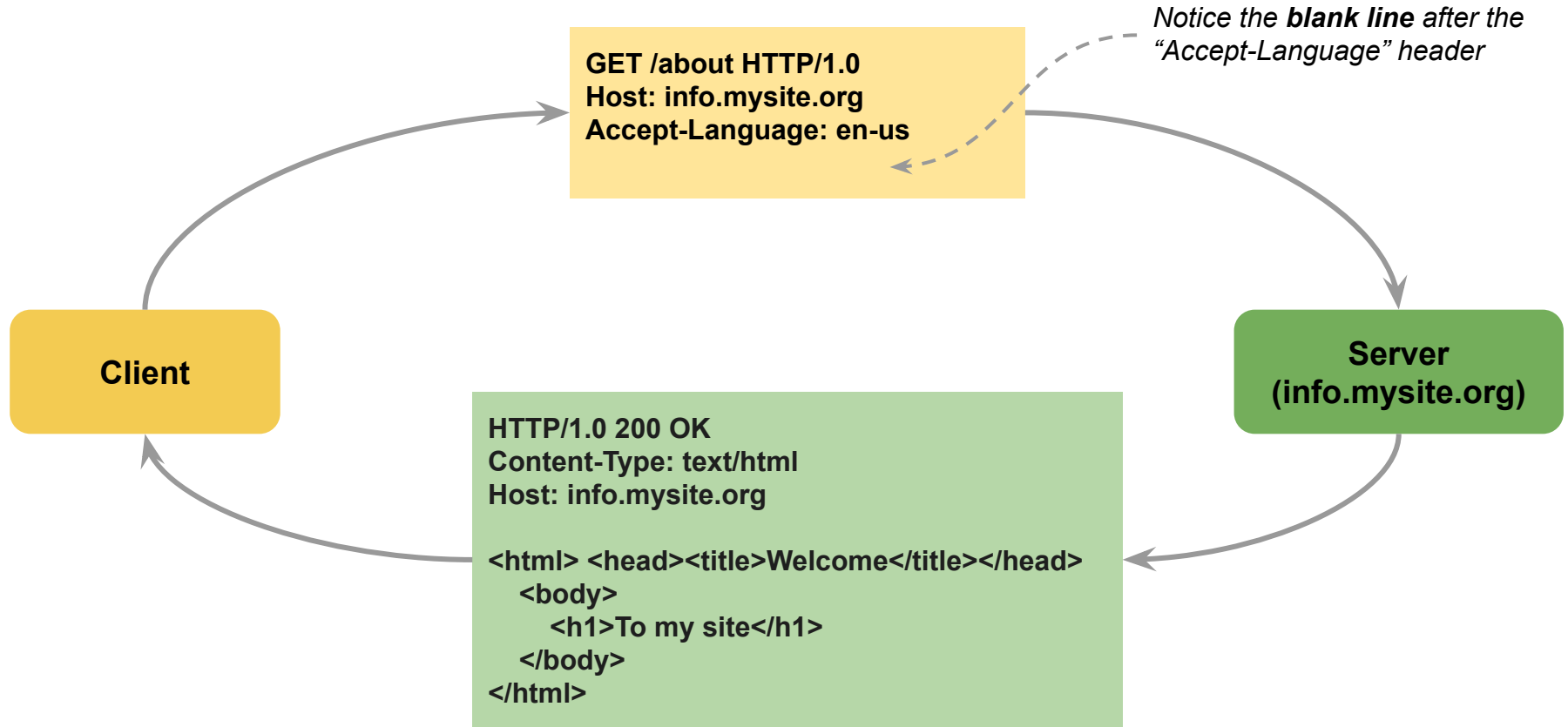
http:// www.gvsu.edu /pcec/advising/ index.html



HTTP Messages: Request & Response

Demo: URL & Web Dev Tools

http://info.mysite.org/about/



Web Browser DevTools (Network Tab)

`http://info.cern.ch`

The screenshot displays the Web Browser DevTools interface with the Network tab selected. The top toolbar includes icons for Inspector, Console, Debugger, and the Network tab, which is highlighted with a red box. Below the toolbar, the 'Filter URLs' input is visible. The main panel shows a list of network requests, with the first request 'GET http://info.cern.ch/' selected and highlighted in blue. To the right of this list, the details for the selected request are shown. The 'Status' is '200 OK' with a question mark icon. The 'Version' is 'HTTP/1.1'. The 'Transferred' size is '646 B (646 B size)'. The 'Request Priority' is 'Highest'. The 'DNS Resolution' is 'System'. Below the status section, the 'Response Headers (232 B)' are listed, with a red arrow pointing to the section header. The headers include: 'Accept-Ranges: bytes', 'Connection: close', 'Content-Length: 646', 'Content-Type: text/html', 'Date: Mon, 11 Sep 2023 14:25:55 GMT', 'ETag: "286-4f1aadb3105c0"', 'Last-Modified: Wed, 05 Feb 2014 16:00:31 GMT', and 'Server: Apache'. Below the response headers, the 'Request Headers (346 B)' are listed, with a red arrow pointing to the section header. The request headers include: 'Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,*/*;q=0.8', 'Accept-Encoding: gzip, deflate', 'Accept-Language: en-US,en;q=0.5', 'Connection: keep-alive', 'Host: info.cern.ch', 'Upgrade-Insecure-Requests: 1', and 'User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:109.0) Gecko/20100101 Firefox/117.0'.

Inspector Console Debugger **Network** Style Editor Performance >>

Filter URLs

All HTML CSS JS XHR Fonts Images Media WS Other

Do File I... 1 T... Size

2 G / doc h c... 646 B

GET http://info.cern.ch/

Filter Headers Block Resend

Status **200 OK** ?

Version HTTP/1.1

Transferred 646 B (646 B size)

Request Priority Highest

DNS Resolution System

Response Headers (232 B) Raw

Accept-Ranges: bytes

Connection: close

Content-Length: 646

Content-Type: text/html

Date: Mon, 11 Sep 2023 14:25:55 GMT

ETag: "286-4f1aadb3105c0"

Last-Modified: Wed, 05 Feb 2014 16:00:31 GMT

Server: Apache

Request Headers (346 B) Raw

Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,*/*;q=0.8

Accept-Encoding: gzip, deflate

Accept-Language: en-US,en;q=0.5

Connection: keep-alive

Host: info.cern.ch

Upgrade-Insecure-Requests: 1

User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:109.0) Gecko/20100101 Firefox/117.0

Fetch the content via the command line

```
curl --verbose http://info.cern.ch
```

(On Linux/OSX/Windows 10 WSL)

Fetch the content via the command line

```
iwr http://info.cern.ch -UseBasicParsing
```

(On Windows PowerShell)

HTTP Request/Response

line

1 Request/Response line

2 Header1: value1

3 Header2: value2

... ... more header lines here ... HeaderN:

N valueN

N+1 One blank line

N+2 message body

N+3 (plain text or binary)

N+4

...

required

Header lines (optional)

required

Message body (optional)

- Data for POST requests, examples
 - Encrypted userid/password
 - Encrypted credit card details
 - Content of uploaded file(s)
 - etc.
- Returned contents of server responses
 - HTML doc
 - Image data
 - etc.

HTTP headers of interest to web developers

Header	Description	Example
Accept	Inform server media-type to respond	Accept: image/jpg
Accept-Language	Inform the server which languages the client is able to understand	Accept-Language: en-US; en-UK
Content-Type	Media type of the returned content	Content-Type: plain/text
Content-Language	The languages of the content	Content-Language: en-US
Date	Date and time of the message	Date: Mon, 21 Aug 2017 18:14:36 GMT
ETag	Identifier used by caching algorithms	ETag: ""8a9-291e721905000"
Host	Specify the domain name of the intended server (mainly for Virtual Hosting)	Host: www.personal.me:5555

HTTP 1.0 Commands (Request Methods)

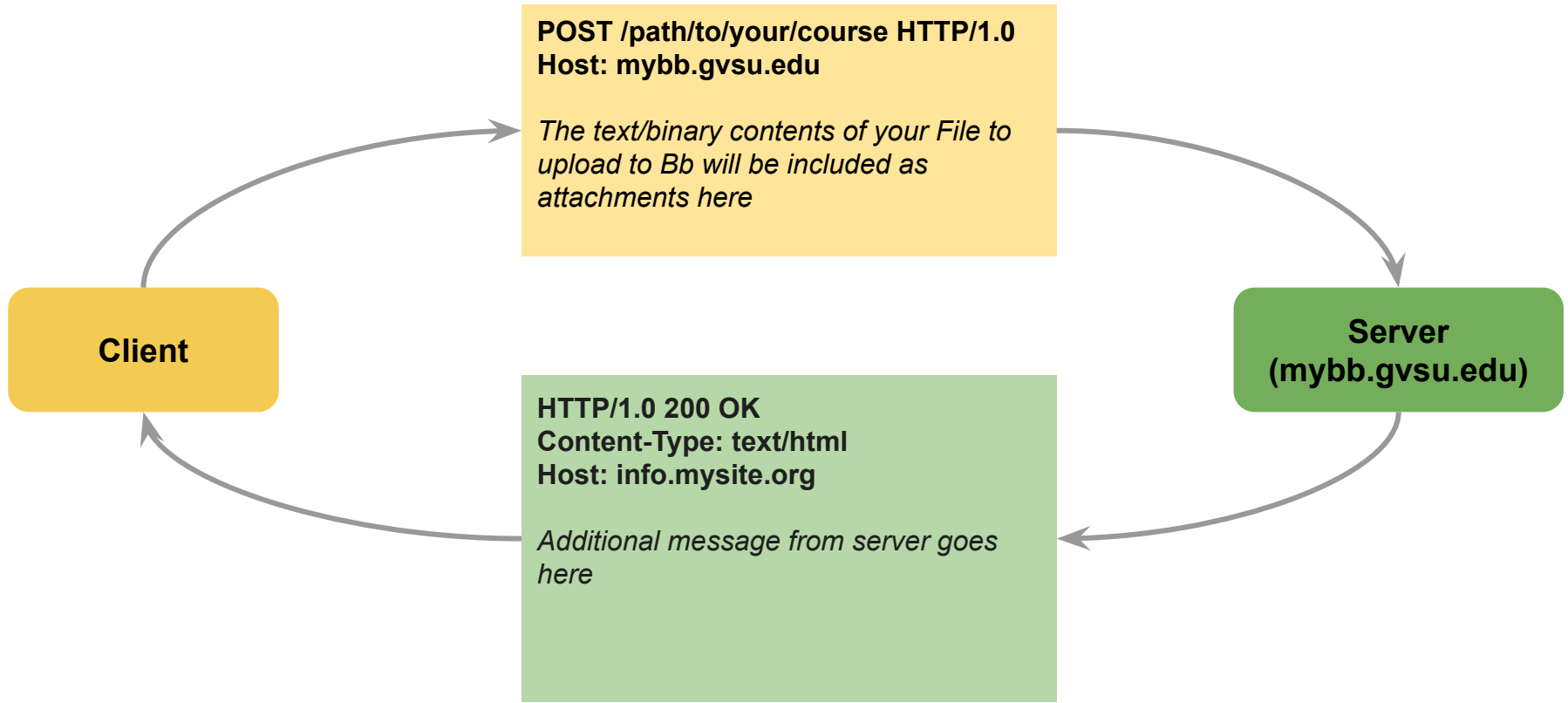
- GET
 - POST
 - HEAD
- 
- More-frequently used

(like GET but the server responds only with header, no data)

- PUT
 - DELETE
 - OPTIONS
- 
- Less-frequently used

Operation	HTTP Request
Create	POST
Read	GET
Update	PUT
Delete	DELETE

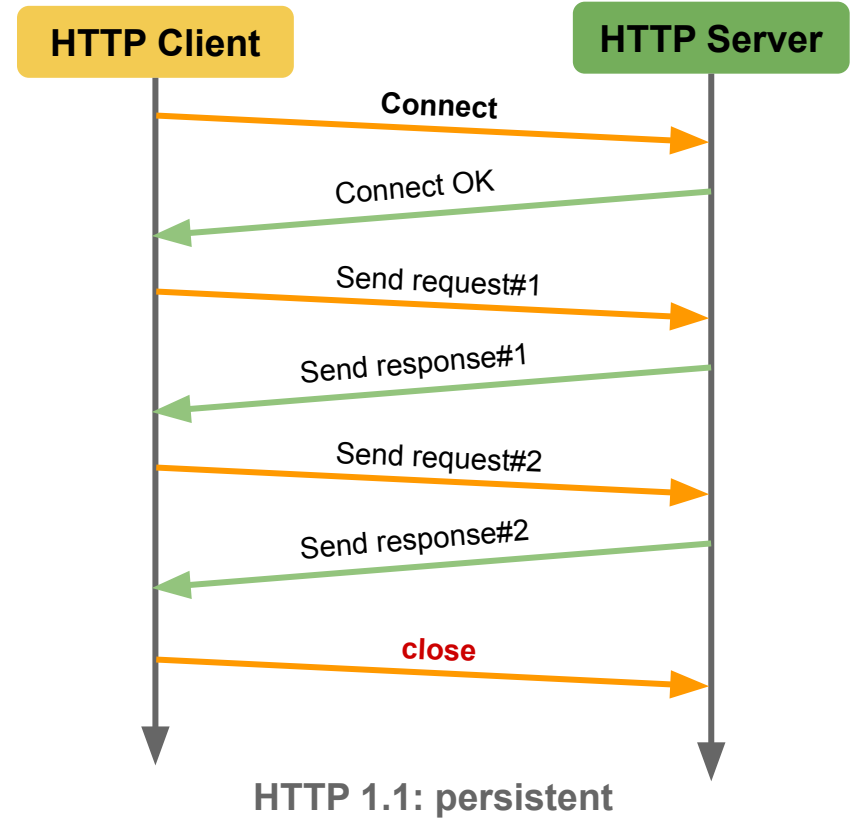
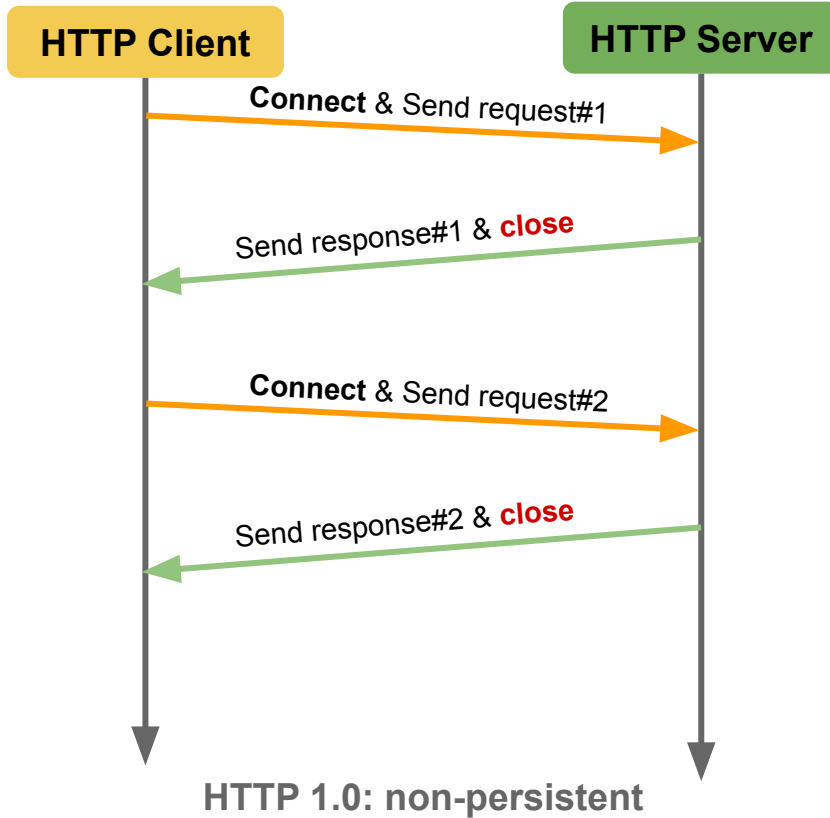
POST: upload file to Bb



HTTP Status Code

Status Code	Description
1xx	Informational messages
2xx	Success messages
3xx	Redirect message
4xx	Error on the client's behalf
5xx	Error on the server's behalf

HTTP Connections: Persistence



HTTP 1.0 vs. HTTP 1.1

HTTP 1.0

- One request per connection (non-persistent)
- Cache control is **timestamp based** with one-second resolution (inaccurate)
- Client cannot request a portion of a resource
- Responses are delivered in one big chunk

HTTP 1.1

- N requests per connection (persistent)
- Response can be delivered in chunk
- Cache control is **content based**, responses include entity tag (Etag), similar to hash value
- Clients can request **partial content**
 - “Range:” header line in HTTP request
- Responses may be delivered in many small chunks

HTTP 1.1 vs. HTTP 2

HTTP 1.1

- HTTP messages encoded in text format
- Require multiple connections to achieve concurrency
- Uncompressed response headers
- No resource prioritization

HTTP 2

- HTTP messages encoded in binary format
 - Message = request or response
- Multiple concurrent channels on a single connection
- Compressed response headers
- Resource prioritization (important requests complete more quickly)